

Task 1: Install Wireshark on your laptop, start a capture on your main network interface, and visit any website (like instagram.com). Stop the capture and note down the protocol names you see in the packet list.

Completed

ip.addr == 192.168.29.7					
No.	Time	Source	Destination	Protocol	Length Info
276	10.740943000	192.168.29.7	65.0.129.216	TLSv1.2	136 Application Data
277	10.741187400	192.168.29.7	65.0.129.216	TLSv1.2	102 Application Data
278	10.741660600	192.168.29.7	65.0.129.216	TLSv1.2	136 Application Data
279	10.741768700	192.168.29.7	65.0.129.216	TLSv1.2	102 Application Data
280	10.757899300	65.0.129.216	192.168.29.7	TCP	54 443 → 56511 [ACK] Seq=1 Ack=131 Win=4 Len=0
281	10.757900400	65.0.129.216	192.168.29.7	TLSv1.2	287 Application Data
282	10.758075000	65.0.129.216	192.168.29.7	TLSv1.2	204 Application Data
283	10.758147700	192.168.29.7	65.0.129.216	TCP	54 56511 → 443 [ACK] Seq=261 Ack=384 Win=255 Len=0
284	10.758962500	192.168.29.7	65.0.129.216	TLSv1.2	136 Application Data
285	10.759151200	192.168.29.7	65.0.129.216	TLSv1.2	102 Application Data
287	10.772990000	65.0.129.216	192.168.29.7	TCP	54 443 → 56511 [ACK] Seq=384 Ack=391 Win=4 Len=0
288	10.774342100	65.0.129.216	192.168.29.7	TLSv1.2	192 Application Data
295	10.803367100	192.168.29.7	65.0.129.216	TLSv1.2	136 Application Data
296	10.803628000	192.168.29.7	65.0.129.216	TLSv1.2	102 Application Data
297	10.804376300	192.168.29.7	65.0.129.216	TLSv1.2	137 Application Data
298	10.804609300	192.168.29.7	65.0.129.216	TLSv1.2	102 Application Data
304	10.815059400	192.168.29.7	65.0.129.216	TLSv1.2	137 Application Data
305	10.815320000	192.168.29.7	65.0.129.216	TLSv1.2	102 Application Data
308	10.821111900	65.0.129.216	192.168.29.7	TCP	54 443 → 56511 [ACK] Seq=522 Ack=652 Win=4 Len=0
309	10.821113200	65.0.129.216	192.168.29.7	TLSv1.2	398 Application Data

Task 2: Pick any 3 packets from your Wireshark capture and identify which OSI layer each packet mainly represents (for example, HTTP for Application layer, TCP for Transport layer, etc.). Hint: Use the Protocol column and Info details in Wireshark to help you map protocols to OSI layers.

OSI Layer: Presentation (Layer 6)

TLS Client Hello + Application Data

192.168.29.7 → 65.0.129.216 TLSv1.2 136 Application Data

OSI Layer: Transport (Layer 4)

TCP SYN/ACK packets

65.0.129.216 → 192.168.29.7 TCP 443 → 56511 [ACK] Seq=1 Ack=131 Win=4 Len=0

OSI Layer: Network (Layer 3)

192.168.29.7 → 224.0.0.22 IGMPv3 Membership Report / Join group

Task 3: Create a colorful diagram of the OSI Model using any drawing tool (MS Paint, Canva, or even paper and upload a photo). For each layer, add a real-world example from apps you use daily (e.g., WhatsApp uses the Application layer, Instagram images use the Presentation layer, etc.).

THE OSI MODEL

7 Layers of networking

Layer 7 - Application

📱 WhatsApp

The Application layer where users interact with network services. This is where apps like WhatsApp operate — sending and receiving chat messages, voice notes, and calls directly over the internet.

Layer 6 - Presentation

📷 Instagram

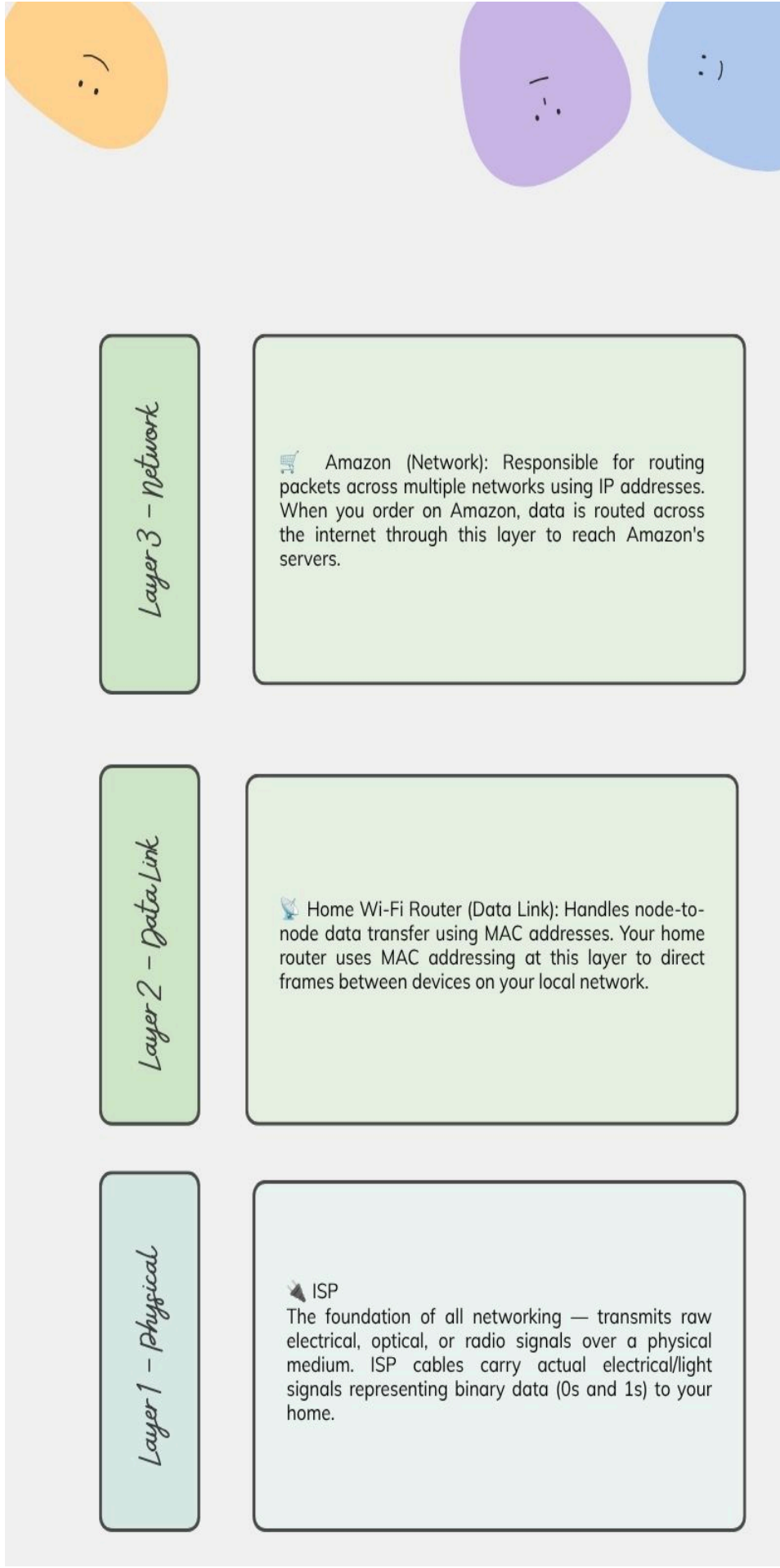
Translates, encrypts, and compresses data. Instagram uses this layer to encode photos into JPEG/PNG formats and apply end-to-end encryption before transmitting your images and stories.

Layer 5 - Session

💻 Zoom (Session): Manages and maintains active communication sessions. Zoom uses this layer to keep your video call session alive, synchronize audio/video streams, and handle reconnections.

Layer 4 - Transport

📡 Paytm (Transport): Ensures reliable, ordered, and error-checked delivery of data between devices. Paytm relies on TCP at this layer for secure, complete transaction data delivery.



Task 4 : Explain how sending a message on WhatsApp travels through at least 4 layers of the OSI Model, describing what happens at each layer in your own words. Hint: Think about how your message is typed, packaged, sent over the internet, and received.

After a message is typed: "How are you?"

Layer 7 — Application

This is where you actually type the message in the WhatsApp app. WhatsApp prepares your text to be sent — this is the layer the user directly interacts with.

Layer 6 — Presentation

Before your message leaves your phone, WhatsApp encrypts it (end-to-end encryption, so no one in between can read it) and converts it into a standard format the receiving phone will understand.

Layer 5 — Session

WhatsApp keeps a connection "session" alive between your phone and WhatsApp's servers — this lets us send several messages back-to-back without reconnecting every single time.

Layer 4 — Transport

Your message gets broken into small chunks called segments. This layer makes sure every chunk arrives, arrives in the right order, and if one gets lost, it's resent. This is why your message doesn't show up with missing words.

Layer 3 — Network

Each segment gets placed into a packet with an IP address stamped on it — the phone's address (sender and receiver) and WhatsApp's server address — so the internet knows where to route it.

Layer 2 — Data link

On the home network, this layer moves the data from your phone to your Wi-Fi router using MAC addresses. It also checks if the data packets are correct.

Layer 1 — Physical

Finally, the message becomes actual signals — radio waves over Wi-Fi from your phone to the router, then light pulses or electrical pulses to pass through cable to your ISP and onward across the internet.

Your friend's phone then does this whole process in reverse — Physical → Data link → Network → Transport → Session → Presentation → Application — to turn those signals back into a message on their screen.